

Sensitivity-Aware Air Purifier

Innovation dossier

1. What is it?

A concept for the real 'Ozone / smell / irritation (weak real signal)' friction, built with the PERSONALISE design move - room/system -> individual/context.

2. Physical architecture

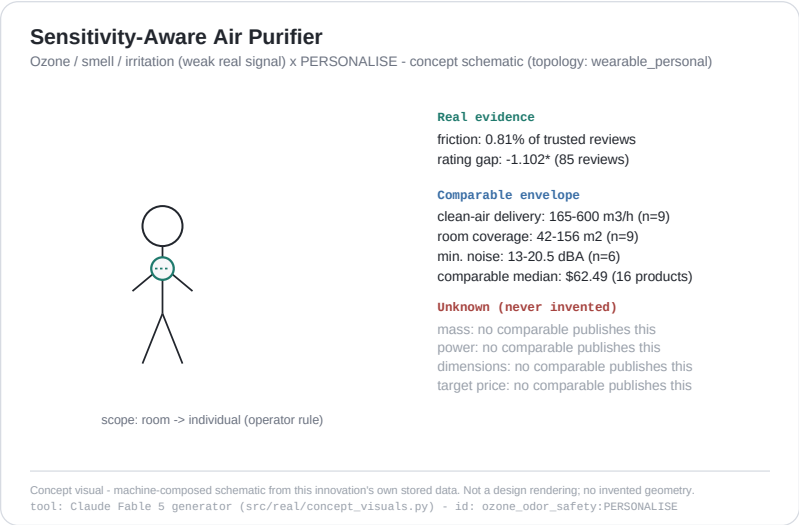


Figure: [Concept visual](#)

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

artifact visual-ozone_odor_safety:PERSONALISE - /concept-visuals/ozone_odor_safety_PERSONALISE.svg

Product archetype

- room scale:** Unknown - no real data in this pipeline for this field.
 - mobility:** Personal/portable - operator PERSONALISE moves scope from room to individual.
 - size class:** Unknown - no real data in this pipeline for this field.
 - power class:** Unknown - no real data in this pipeline for this field.
 - noise ambition:** Unknown - no real data in this pipeline for this field.
 - maintenance burden:** Unknown - no real data in this pipeline for this field.
 - filter service dependency:** Unknown - no real data in this pipeline for this field.
 - connected:** Unknown - no real data in this pipeline for this field.
 - form factor:** wearable_personal
- Design rule - each populated field follows from the operator's own fixed definition, never from evidence

System architecture

Unknown - no system-architecture dataset exists; the archetype above is a design rule from the operator's own definition.

Method choice

PERSONALISE - Room/system -> individual/context.

Method choice - authored design vocabulary

3. Why it exists

Reality Observed

0.81% of trusted reviews (85 reviews across 29 real products) carry this friction, with a -1.102* average rating gap. Signal: Ozone/smell/irritation complaints converge with real regulatory guidance on ozone safety limits.

Households whose real reviews carry this friction: 85 reviews across 29 real products (82.4% verified purchase, 2011-05-18-2023-04-13).

Unknown - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

Transformation Method choice

METHOD CHOICE, not evidence: the authored operator table assigns PERSONALISE - "Room/system -> individual/context." - to this friction. Analyst design judgment, applied deterministically.

Product consequence Feasibility precedent

Feasibility precedent (from the trend corpus, not a consumer-evidence consequence): Real, directly-relevant regulatory and technical precedent exists: TC-R04 (CARB's real certified-device list is the existing regulatory bar for ozone-safety compliance sensor-led guidance would need to respect) and TC-R08 (Matter's real already-standardized Air Quality Sensor cluster could carry a placement/sensitivity signal without a new protocol).

4. Evidence and lineage

Parent paths

No parent paths - this innovation cites no tension or assumption paths; its grounding is the friction evidence in section 3, why it exists.

evidence ids: taxonomy:ozone_odor_safety, TC-R04, TC-R08

category assumptions: No category assumption shares a paper with this theme's signal evidence.

Design DNA

F consumer friction (taxonomy:ozone_odor_safety) Present
Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator Method choice
Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (ozone_odor_safety) Present
Ozone/smell/irritation complaints converge with real regulatory guidance on ozone safety limits.

T scientific tension Missing / unverified
No real research tension shares a paper with this theme's signal evidence.

R rival gap Missing / unverified
No real competitor-weakness data clears the white-space threshold for this theme.

C versuni capability Missing / unverified
No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption Missing / unverified
No category assumption shares a paper with this theme's signal evidence.

E economic condition Present
Real price-weighted exposure computed for this theme (\$16,112.87).

5. Comparable products

comparable set: 10 verified official Versuni/Philips air-purifier families (room-scale)
The operator implies a different form factor than the room-scale verified comparables - the ranges above are category context, not a target for this concept.

clean-air delivery rate (CADR): 165-600 m3/h across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

room coverage: 42-156 m2 across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

minimum noise: 13-20.5 dBA across 6 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

6. Engineering envelope

metric	value from verified comparables	epistemic type
clean-air delivery rate (CADR)	165-600 m3/h (n=9 comparables)	Observed comparable
room coverage	42-156 m2 (n=9 comparables)	Observed comparable
minimum noise	13-20.5 dBA (n=6 comparables)	Observed comparable
target mass	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target power	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	Unknown

7. Economics

price-weighted exposure: \$16112.87

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$62.49 (n=16 products) - not this concept's price

MEDIAN real listed price across the 16 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

target price: Not modelled - No concept target-price model exists - only the comparable market median (\$62.49, 16 real products) as reference. A target price would be invented.

8. Prototype

State: [Concept visual](#)

Concept visual means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

9. Experiment and next learning

Next experiment

Confirmation that the relevant Versuni product lines are already CARB-certified would further strengthen this; the reverse would lower it.

Kill criterion

No authored kill criterion yet for this innovation - only the formal case (see formal_case.abandon_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

10. What could kill it?

Where this idea stands: rejected - graveyard

Killed in the Magic funnel: Another concept beats it on pain, value, and feasibility.

kill criterion: No authored kill criterion yet for this innovation - only the formal case (see formal_case.abandon_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

Uncertainties

- No real Versuni data on current CARB-certification status of the specific product lines this would ship on.

Critic dimensions

Evidence	Challenge	Gate passed but prevalence (0.81%) is only marginally above the materiality floor - thin evidence base.
Economic	Survive	Real price-weighted exposure (\$16,113) is at or above the corpus median (\$16,113).
Competitive	Needs evidence	No real rival-weakness data recorded for this friction theme.
Human	Needs evidence	No real usability/human-factors evidence exists in this pipeline for this concept.
Physical	Needs evidence	No real lab/engineering feasibility data exists in this pipeline for this concept.
Versuni fit	Needs evidence	No real organizational-capability assessment exists in this pipeline for this concept.
Timing	Needs evidence	No real market-timing/window evidence exists in this pipeline for this concept.
Robustness	Needs evidence	No real sensitivity/robustness study exists in this pipeline for this concept.

critic overall: Challenge

Criteria results

Needs evidence: 35 - Pass: 10 - Challenge: 3 - N/A: 3 - Kill: 1

V8	Challenge	Operator PERSONALISE may add surface area to manage - unverified.
C1	Kill	Not white space.
B2	Challenge	Sensor/automation-dependent theme - see RP-09/RP-10 precision-vs-accuracy tension.
F4	Challenge	ozone_odor_safety theme - real regulatory scrutiny (TC-R02, TC-R04) applies.

Lineage and provenance

generated by: src/real/innovations_real.py
magic run input sha256: 0b3f9862e0e270a295eacd709182802e981090e817f4f1072000e37aa83f150c
funnel snapshot: d5d03b0d3856721f04d7d1c4ee7f2fe2a7d1df64ac4d2d4130bede07d2b13f61
figure source: /concept-visuals/ozone_odor_safety_PERSONALISE.svg (generated 2026-08-31T18:00:05.654587+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)
data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)
dossier generator: src/real/generate_innovation_dossiers.py